

MERA: A Green Edge Resource Control System With Privacy-Preservation via Mean-Field Reinforcement Learning

Ziqi Wang¹, Xiaoyu Xia¹, Senior Member, IEEE, Ibrahim Khalil², Tianxu Lan, Feng Liu,
Xiaolong Xu¹, Senior Member, IEEE, Xun Yi¹, Minhui Xue¹, Senior Member, IEEE,
and Elisa Bertino³, Fellow, IEEE

Abstract—The global rollout of 5G networks has spurred the rapid deployments of edge servers for hosting latency-sensitive web applications, which improves quality of experience (QoE). However, current efforts fall short in the substantial energy costs associated with the 24/7 operation of edge servers and overlook user privacy by requiring accurate user information for service provision, eroding the sustainability of multi-access edge computing (MEC). To enhance the QoE and service performance while ensuring privacy in MEC, we systematically formulate the interaction among edge servers as a privacy-preserving experience-aware edge resource control (PEERC) problem. To address this, we conduct a global resource control and propose a collaborative resource allocation system named MERA. MERA leverages k -anonymity data obfuscation to protect user location and resource demand privacy while enhancing service performance and energy efficiency with mean-field multi-agent reinforcement learning. Extensive experiments based on a synthetic real-world dataset demonstrate that MERA significantly surpasses benchmarks in terms of QoE, user coverage, privacy, and energy efficiency by $1.18\times$, $1.24\times$, $1.63\times$, and $1.27\times$ on average.

Index Terms—Edge resource control, edge privacy protection, k -anonymity, mean-field control, multi-agent reinforcement learning.

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Ziqi Wang, Xiaoyu Xia, Ibrahim Khalil, and Xun Yi are with the School of Computing Technologies, RMIT University, Melbourne, VIC 3000, Australia (e-mail: ziqi.wang6@student.rmit.edu.au; xiaoyu.xia@rmit.edu.au; ibrahim.khalil@rmit.edu.au; xun.yi@rmit.edu.au).

Tianxu Lan is with the Department of Mathematics, The Chinese University of Hong Kong, Hong Kong (e-mail: 1155184513@link.cuhk.edu.hk).

Feng Liu is with the School of Computing and Information Systems, The University of Melbourne, Melbourne, VIC 3010, Australia (e-mail: feng.liu1@unimelb.edu.au).

Xiaolong Xu is with the School of Computer and Software, Nanjing University of Information Science and Technology, Nanjing 210044, China (e-mail: njxlu@gmail.com).

Minhui Xue is with the CSIRO's Data61 and Responsible AI Research (RAIR) Centre, Eveleigh, NSW 2015, Australia (e-mail: jason.xue@data61.csiro.au).

Elisa Bertino is with the Department of Computer Science, Purdue University, West Lafayette, IN 47907-2107 USA (e-mail: bertino@purdue.edu).

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I. INTRODUCTION

THE rapid growth of data-intensive applications and the increasing demand for low-latency services have made multi-access edge computing (MEC) a pivotal component of modern computing paradigms [1], [2]. By redistributing resources to the network edge, MEC offloads the computing burden of mobile devices and enables real-time web services to improve users' quality of experience (QoE). The advancement of MEC systems has fueled a wide range of research across domains, including edge services [3], [4], edge intelligence [5], [6], and edge privacy and security [7], [8].

Allocating edge server resources to fulfill users' computing demands is a core process in MEC environments, particularly for ensuring high quality of service (QoS). MEC systems dynamically distribute resources in real time based on user requests, a process known as edge demand response (EDR). However, EDR has raised increasing concerns due to the growing energy consumption of data centers, which may account for up to 8% of global electricity by 2030 [9]. As an extension of cloud computing, MEC further introduces dense deployments of edge servers near base stations, with 5G systems potentially reaching up to 50 servers per square kilometer [10]. This high density significantly amplifies energy consumption, leading to concerns about environmental sustainability and operational costs. Recent efforts attempt to mitigate this issue by adopting strategies such as workload shifting and server consolidation [11], [12], where unnecessary servers are turned off to reduce energy usage. However, such energy-saving strategies often rely on accurate user data (e.g., locations and service requests), which contain sensitive information and raise serious privacy concerns. From a user perspective, sacrificing privacy for energy efficiency offers limited incentives and may hinder the adoption of edge services, making privacy preservation essential for sustainable MEC systems.

As a key factor in web services, QoE measures user satisfaction by combining system performance with user perception, including responsiveness and reliability [13]. For example, in streaming platforms, QoE reflects real-time responsiveness for applications such as gaming and video delivery, directly shaping user experience. In MEC environments, QoE can be improved by allocating sufficient edge resources (e.g., bandwidth and

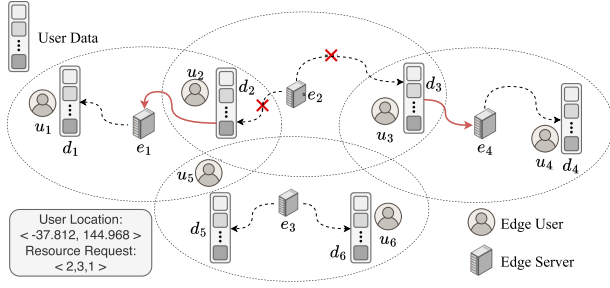


Fig. 1. An MEC environment example. Initial allocation keeps all servers active; optimized reallocation powers off e_2 by reassigning u_2, u_3 , saving energy while maintaining QoE.

memory) to users. Maintaining high QoE is essential for user retention and service success. However, achieving this remains challenging due to dynamic demand, as well as the need to balance privacy protection and energy efficiency with service performance.

Fig. 1 illustrates a MEC system with four edge servers $\{e_1, \dots, e_4\}$ and six users $\{u_1, \dots, u_6\}$ accessing web services with sensitive data. Each server provides multi-dimensional resources (e.g., bandwidth, GPU, memory) to support nearby users. Based on request data $\{d_1, \dots, d_6\}$, including user location and resource demands, servers determine allocation strategies for service delivery. For example, users u_1 and u_4 are assigned to e_1 and e_4 , respectively. However, the continuous operation of semi-trusted edge servers raises concerns regarding unnecessary energy consumption and potential privacy leakage.

To improve energy efficiency, redundant servers can be powered off through optimized allocation. For example, reallocating users u_2 and u_3 to e_1 and e_4 allows e_2 to be shut down. However, incorporating privacy protection further complicates the problem and may degrade system performance. Existing techniques, such as homomorphic encryption [14] and differential privacy [15], provide strong guarantees but often introduce noise, distortion, and latency, making them unsuitable for real-time MEC services. This can lead to inaccurate resource allocation and reduced QoE, coverage, and energy efficiency. Therefore, achieving both efficiency and privacy remains a critical challenge.

Recently, k -anonymity has gained significant attention in data privacy protection due to its balance between privacy and data utility [16]. Unlike differential privacy, which often introduces substantial noise to protect individual data points, k -anonymity ensures that each user is indistinguishable from at least $k - 1$ other users, which minimizes data distortion, preserving data accuracy while still maintaining privacy. Compared to existing approaches, k -anonymity strikes a balance between privacy and data utility, making it more practical for efficient and high-performance edge services. Moreover, in the EDR process, the server-server interactions during the resource allocation process could be further modeled as a multi-agent system to enable collaborative resource control from a global perspective. Each edge server acts as an agent, making decisions in a distributed manner while cooperating with others to optimize resource allocation across the network. This multi-agent system approach

allows for more flexible and adaptive resource management that improves overall system performance from a globally collaborative perspective.

To tackle the privacy-preserving experience-aware edge resource control (PEERC) problem, we propose a novel **mean-field edge resource allocation (MERA)** framework. MERA protects user privacy via k -anonymity while enhancing overall system welfare, including service quality and energy efficiency, through mean-field multi-agent reinforcement learning (MARL). Unlike existing approaches, MERA provides a *unified, co-designed framework* that tightly couples privacy-aware state abstraction with scalable multi-agent decision-making. This design enables a balanced trade-off between user privacy and system performance, ensuring accurate resource allocation while avoiding unnecessary server power-offs. The contributions of this paper are summarized as follows:

- To the best of our knowledge, this is the first work to systematically formulate the *privacy-preserving experience-aware edge resource control* (PEERC) problem, providing a unified perspective that jointly enhances user privacy, QoE, and energy efficiency in large-scale MEC systems.
- To address this issue, we propose MERA, a *co-designed framework* that incorporates k -anonymity with mean-field MARL. Unlike prior approaches, we jointly embed the privacy preservation into the system state and directly influence the decision-making process.
- We introduce a mean-field approximation that transforms complex multi-agent interactions into a scalable collective state, improving edge scalability and stability.
- We further develop a mean-field resource Q-control algorithm that operates under the privacy-aware and mean-field-coupled environment to optimize global resource allocation.
- Extensive experiments and ablation studies demonstrate that MERA consistently outperforms state-of-the-art benchmarks, achieving superior trade-offs among user privacy and system performance.

II. PRELIMINARIES

A. K -Anonymity

As a privacy-preserving approach, k -anonymity was developed to protect individual identities in datasets by ensuring that each individual cannot be distinguished from at least $k - 1$ other entries [17]. It has been widely applied in data privacy protection, especially in cases where sensitive personal information like medical or financial records is shared, which can be represented by:

Definition 1 (*k -anonymity data protection*): Let $d_j = (x_j, y_j)$ denote the real data of user u_j , and let $\mathcal{D} = \{d_1, d_2, \dots, d_J\}$ be the set of all user data. A user's data satisfies k -anonymity if there exists an anonymized set $\mathcal{D}' \subseteq \mathcal{D}$ with $|\mathcal{D}'| = k$ and $d_j \in \mathcal{D}'$, such that the probability of correctly identifying d_j from $\mathcal{D}'$ is at most $\frac{1}{k}$:

$$Pr(d_j | \mathcal{D}') \leq \frac{1}{k}, \quad \forall d_j \in \mathcal{D}'. \quad (1)$$

where $\mathcal{D}'$ denotes the anonymized candidate set containing k indistinguishable instances, $Pr(d_j | \mathcal{D}')$ denotes the probability of identifying the true data d_j within the set $\mathcal{D}'$.

Practically, privacy mechanisms in MEC systems should preserve service quality while introducing minimal overhead. Although methods such as differential privacy (DP) provide strong theoretical guarantees, their noise injection may seriously distort resource demand patterns and affect real-time decision making. In contrast, k -anonymity preserves the relative structure of user requests while providing indistinguishability among users, better aligning with latency-sensitive and resource-aware MEC environments. Moreover, its privacy level can be directly controlled by the parameter k , making it more interpretable and easier to tune in practice.

B. Mean-Field Control

Mean-field control theory has gained attention for optimizing large-scale multi-agent systems [18], [19]. Originating from stochastic processes and partial differential equations, it models the limiting behavior of large populations of interacting agents. Unlike traditional control methods focused on individual dynamics, mean-field control captures global statistical interactions, reducing complexity and improving computational efficiency.

The McKean-Vlasov stochastic differential equation (SDE) in mean-field control describes the evolution of a representative agent's state, influenced by the population's empirical distribution:

Definition 2 (McKean-Vlasov Process): Given $\{b_t, \sigma_t\} : \mathbb{R} \times \mathcal{P}(\mathbb{R}) \rightarrow \mathbb{R}$, where $\mathcal{P}(\mathbb{R})$ is the space of probability measure on $\mathbb{R}$. A stochastic process $(X_t)_{t \geq 0}$ is a McKean-Vlasov process if it solves:

$$dX_t = \sigma_t^2(X_t, \mu_t)dt + b_t(X_t, \mu_t)dW_t \quad (2)$$

where μ_t describes the distribution of X_t and W_t denotes a standard Brownian Motion.

This describes the user-server states in MEC and models the remaining resources of edge machines in Section III. More details of the mean-field approach will be discussed in Section IV-B.

III. PROBLEM FORMULATION

In this section, we introduce the system model, potential adversaries, and system factors to formulate the privacy-preserving experience-aware edge resource control (PEERC) problem. We've provided the core notations in Table I, and detailed notations can be found in Table VI.

A. System Model

Consider J users $\{u_1, \dots, u_J\} \in \mathcal{U}$ sending b -dimensional computing requests $\{y_1, \dots, y_J\} \in \mathcal{Y}$, ($y_j = \{y_j^1, \dots, y_j^b\}$), such as bandwidth, GPU, memory, etc., and H edge machines $\{e_1, \dots, e_H\} \in \mathcal{E}$ with c -dimensional computing resources $\{z_1, \dots, z_H\} \in \mathcal{Z}$, ($z_h = \{z_h^1, \dots, z_h^c\}$) within a MEC system. Each edge machine covers a specific geographical area with a

TABLE I
KEY NOTATIONS

System and Threat Model	
$\{u_1, \dots, u_J\} \in \mathcal{U}$	mobile users
$\{y_1, \dots, y_J\} \in \mathcal{Y}$	users' computing requests
x_j^t	location of user u_j at time t
$d_j^t = (x_j^t, y_j^t)$	user request tuple
$\{e_1, \dots, e_H\} \in \mathcal{E}$	edge servers
H	number of edge servers
$p_h(t)$	remaining resources of server e_h
$\mathcal{K}(u_i)$	neighboring servers of user u_i
$\mathcal{A}^t$	resource allocation decisions at time t
$\mathcal{G}$	data indistinguishability mechanism
$\mathcal{T}(\cdot)$	adversarial inference function
$\mathcal{R}_j$	privacy level of user u_j
Control and Objectives	
π	resource control policy
Q	Q-function
θ^p, θ^c	actor and critic parameters
α	learning rate
β	exploration rate
s	system state
$\Upsilon_c(\mathcal{A}^t)$	user coverage
$\Upsilon_p(\mathcal{A}^t)$	system privacy
$\Upsilon_e(\mathcal{A}^t)$	energy efficiency
$\eta(a_i^t)$	QoE of user u_i
ι	QoE threshold
w	weighting coefficients

radius of $\{f_1, \dots, f_H\}$. Given the characteristics of randomness in both the number of user requests and resource amount in MEC environments, the interval of a resource request proposed by user u_j follows a domain-specific distribution, e.g., exponential distribution [20], with mean $1/\lambda$ over time. If a user u_j proposes a resource request at time t , her requested resource amount generally follows the Gaussian distribution:

$$y_j^t \sim N((\mu_j^1, \sigma_j^1), \dots, (\mu_j^b, \sigma_j^b)), \forall t \in [0, T] \quad (3)$$

where μ_j and σ_j are constants denoting the user-specific mean and standard deviation of the b -dimensional resource request of u_j .

Definition 3 (Maximum Service Capacity): Given users whose actual locations x_j^t fall within the coverage radius of edge machine e_h , these users that can be covered by edge server e_h are defined as e_h 's maximum service capacity at time t .

$$\mathcal{N}^t(e_h) = \{u_j | \text{dist}(x_j^t, e_h) \leq \text{cov}_h\} \quad (4)$$

The number of user requests received by the edge machine e_h at time t with the interval of $[t, t+dt)$ is:

$$\sum_{u_j \in \mathcal{N}^t(e_h)} \lambda_j dt + o(dt) \quad (5)$$

where $o(dt)$ represents the degree of magnitude. The total amount of resource requests received by e_h is as:

$$\sum_{u_j \in \mathcal{N}^t(e_h)} \lambda_j \cdot y_j^t dt + o(dt) \quad (6)$$

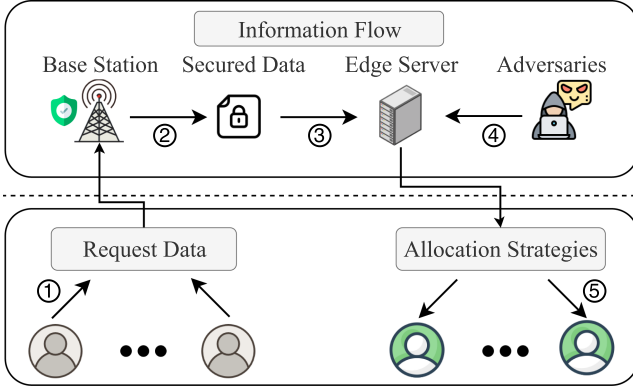


Fig. 2. Edge privacy game. Users send requests to a trusted base station for k -anonymity obfuscation, then to semi-trusted edge servers for collaborative allocation. Adversaries attempt inference on obfuscated data $G(\cdot)$ but are limited by indistinguishability, protecting location/resource privacy in EDR.

Therefore, we consider a non-stationary time-varying EDR model to theoretically describe the dynamics of remaining resources of edge machine e_h at t , represented by the following stochastic differential equation (SDE):

$$-dp_h(t) = \sum_{u_j \in \mathcal{N}^t(e_h)} a_j(t, \hat{p}_H^t) \mu_j \lambda_j dt + a_j(t, \hat{p}_H^t) \lambda_j \sigma_j dW_t \quad (7)$$

where $p_h(t)$ denotes e_h 's remaining resources at time t . $a_j(\cdot)$ is the mean-field control strategy representing the resource proportion of user u_j 's requests satisfied, depending on t , $p_h(t)$, and the amounts of user resource requests. $\hat{p}_H^t = \frac{1}{H} \sum_{e_h \in \mathcal{E}} \delta_{p_h^t}$ is a Dirac measure that denotes the population of remaining resources of edge machines, $\delta_{p_h^t}$ denotes the Dirac delta function.

B. Threat Model

We consider a semi-honest setting in the MEC system, where honest-but-curious edge servers follow the prescribed protocols to serve users faithfully but may attempt to infer users' accurate information from observed data to improve service performance and reduce energy consumption.

We focus on a typical real-world deployment scenario where edge servers operate independently without explicit collusion, and no active adversarial manipulation (e.g., data injection or protocol deviation) is considered. A trusted base station that is managed by the network provider performs data obfuscation before forwarding user requests to edge servers.

C. Potential Adversaries

Users' privacy is a critical concern in MEC systems, where sensitive information such as location and resource demands is continuously exchanged. In this setting, edge servers require user data to enable efficient resource allocation and energy management, while users seek to protect their private information. This interaction forms a privacy-utility trade-off, which can be modeled as a privacy game, as illustrated in Fig. 2. During the EDR process, users first send their requests to a trusted base station, which performs encryption and obfuscation before forwarding

the processed data to semi-trusted edge servers. These edge servers collaboratively perform global resource control based on the obfuscated data. Although adversaries may attempt to infer user information from the received data, the obfuscation mechanism limits the exposure of sensitive information.

User-specific Data Indistinguishability: The ideal way to protect user data is to obfuscate it at a trusted central base station before sending it to the honest-but-curious edge servers for processing, ensuring absolute privacy. The obfuscation mechanism $\mathcal{G}$ can be represented as:

$$\mathcal{G}((x_j^*, y_j^*) | (x_j, y_j)) = \Pr\{x_j^* \in \mathcal{X} | x_j \in \mathcal{X}, y_j \in \mathcal{Y}\} \geq 0 \quad (8)$$

where the probability sum of obfuscated locations is as $\sum_{x_j^* \in \mathcal{X}} \mathcal{G}(x_j^*, y_j^*) | (x_j, y_j) = 1$. In this manner, the data-indistinguishable mechanism $\mathcal{G}$ effectively protects users' information when requesting resources from the edge.

Inferences out of curiosity: In MEC, while malicious adversaries may not be present, *honest-but-curious* edge servers often seek to infer users' actual locations and resource demands to enhance service quality and minimize energy consumption. They launch these potential inferences $\mathcal{J}((\hat{x}_j, \hat{y}_j) | (x_j^*, y_j^*))$ based on the prior knowledge (χ_l, χ_r) of historical and uploaded user locations and computational requests, where $\chi_l(x_j) = \Pr\{x_j \in \mathcal{X}\}$ and $\chi_r(y_j) \sim N((\mu_j^1, \sigma_j^1), \dots, (\mu_j^b, \sigma_j^b))$. By continuously observing the information from users and the prior knowledge, mechanism $\mathcal{J}$ generates estimated user data according to the probability distribution:

$$\mathcal{J}((\hat{x}_j, \hat{y}_j) | (x_j^*, y_j^*)) = \Pr\{x_j \in \mathcal{X}, \hat{y}_j \in \mathcal{Y} | x_j^* \in \mathcal{X}, y_j^* \in \mathcal{Y}\} \geq 0 \quad (9)$$

Representing how distorted the user data is restored, the inference error ψ_j is the expectation of Euclidean distance between x_j^{t*}, y_j^{t*} and inferred location $\hat{x}_j^t \in \mathcal{X}$ with inferred requests $\hat{y}_j^t \in \mathcal{Y}$:

$$\psi_j^l = \sum_{x_j \in \mathcal{X}} \mathcal{J}(\hat{x}_j | x_j^*) \cdot \|x_j - \hat{x}_j\| \text{ and } \psi_j^r = \left\| \frac{y_j^{t*} - y_j}{y_j} \right\|_2^2 \quad (10)$$

More intuitively, a user's privacy $\mathcal{R}$ can be represented by the distortion between their obfuscated data and the real data. As shown in Fig. 3(a), users' privacy of location data is quantified by the ratio of inference error area to the coverage of assigned edge machine e_h . In Fig. 3(b), the privacy of resource data is measured by the l_2 -norm distance during obfuscation. Therefore, individual user privacy could be defined by weighted averaging of these two parts, i.e., $\mathcal{R}_j = w_1 \cdot \left\| \frac{\psi_j^l}{\psi_j^r} \right\|_2^2 + w_2 \cdot \psi_j^r$.

In that case, the overall system privacy $\Upsilon_p(\mathcal{A})$ can be calculated by all users' average privacy value, denoted by:

$$\Upsilon_p(\mathcal{A}^t) = \frac{1}{\mathcal{T}} \cdot \sum_{u_j \in \Delta I, t \in \mathcal{T}} \frac{\mathcal{R}_j}{\mathcal{M}(\mathcal{A}^t)} \in [0, 1] \quad (11)$$

where $\mathcal{M}(\mathcal{A}^t)$ denotes the effective users that are covered by the allocation strategies $\mathcal{A}^t$.

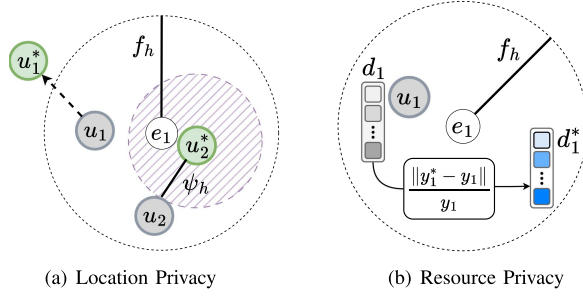


Fig. 3. Definitions of effective user and privacy area (a) Shaded area represents location privacy; (b) Pre/post-obfuscation ratio represents resource privacy.

D. Edge Service Performance

Maintaining optimal system performance in MEC requires carefully considering three key factors: user coverage, quality of experience (QoE), and energy efficiency.

User Coverage: One of the main objectives of MEC systems is to maximize the number of users they can serve within close proximity. To determine the user coverage, two definitions are introduced:

Definition 4 (Adjacent Servers): The adjacent servers for a specific user e_h with her released location x_j^* , the neighboring servers $\mathcal{K}(u_j)$ that cover x_j^* for resource allocation is obtained with:

$$\mathcal{K}(e_h) = \{e_h | \text{dist}(x_j^*, e_h) \leq \text{cov}_h\} \quad (12)$$

Definition 5 (Effective User Set): Consider users with access to edge web services, where both their actual and obfuscated locations are within the coverage of an active edge machine. The set of these effective users is represented by:

$$\Delta I = \{u_j | \text{dist}(x_j^*, e_h) \leq \text{cov}_h \wedge \text{dist}(x_j, e_h) \leq \text{cov}_h\} \quad (13)$$

For instance, u_2 is included as an effective user in Fig. 3(a) as both its real location and obfuscated location are within the coverage of its corresponding server e_1 , while u_1 is not.

Let $a_j = (a_j^s, a_j^r)$ represents the resource allocation strategy, where $a_j^s \in \{0\} \cup \{h | h \in \mathcal{K}(e_h)\}$ ($a_j^s = h$ indicates that user u_j is served by edge machine e_h , while $a_j^s = 0$ means that u_j is not responded by current server. a_j^r is the proportion of resources allocated to users for demand response, where $a_j^r = 1$ means the request is fully responded and $a_j^r = 0$ represents no response.

Let $\mathcal{F}_h(\mathcal{A}^t)$ denote the number of effective users served by edge machine e_h according to the resource allocation decisions. Then, the average number of users covered by $\mathcal{A}^t$ over time is:

$$\mathcal{F}_{total}(\mathcal{A}) = \frac{1}{T} \cdot \sum_{e_h \in \mathcal{E}, t \in T} \mathcal{F}_h(\mathcal{A}^t), \text{ where } \eta_j(a_j^t) \geq \iota \quad (14)$$

where ι is the QoE budget to ensure system effectiveness. Users with QoE higher than ι would be considered as effectively served users and be counted for covered here. The impact on ι could be found as a case study in Section VII.

Here, we use the average user coverage Υ_c in $[0, T]$ to measure the system performance in terms of user coverage:

$$\Upsilon_c(\mathcal{A}) = \frac{\mathcal{F}_{total}}{J} \in [0, 1] \quad (15)$$

Quality of Experience: A service provider aims to maximize user satisfaction by enhancing the quality of service (QoS). An increase in service quality typically leads to higher user satisfaction, i.e., a higher QoE. In edge scenarios, the QoE calculation varies based on the service type. To ensure flexibility and broad applicability, we adopt a general approach [21] to calculate the QoE level of user u_j by:

$$\eta_j(a_j^t) = \frac{\zeta}{1 + e^{-\xi(i(a_j^t) - \gamma)}} \quad (16)$$

to represent a nonlinear correlation between QoS and QoE, where a_j^t is the resource allocation decision of u_j in time slot t , ζ is the maximum experience level, $\xi \geq 0$ controls the increment ratio of the QoE level, and γ determines the mid-point of the QoE function. Meanwhile, the service quality can be derived by:

$$i(a_j^t) = \sup_h \frac{1}{1 - a_j^t(p_h, x_j)} - \frac{1}{a_j^t(p_h, x_j)} \quad (17)$$

which models the impact of resource allocation and QoE. Specifically, when the service quality improves to a certain level, the QoE exhibits diminishing returns with a certain threshold [21]. Please note that various QoE calculations can easily be adjusted in the system by replacing $\eta_j(a_j^t)$ without impacting the model and solution provided in this study.

Edge Energy Efficiency: EDR seeks to minimize edge servers' energy consumption while still meeting nearby users' needs. However, additional energy may be required when location obfuscation is introduced for privacy protection, complicating the PEERC problem. The challenge, therefore, lies in balancing energy efficiency with privacy and utility considerations.

Following the analysis in [22], [23], the total energy consumption can be broken down into three main components: running cost q_h^r , mode switching cost q_h^s , and maintenance cost q_h^m . Particularly, the running cost is represented by:

$$q_h^r(a_j^t) = \sum_{u_j \in \Delta I} \omega_j^r \cdot y_j \cdot a_j^t \quad (18)$$

where ω_j^r is the cost of energy consumption for serving u_j . The mode-switching cost is calculated with $q_h^s = \omega^s(1 - \kappa_r)$, where ω^s is the cost of start-up energy consumption for activating server e_h , and a binary variable $\kappa_j \in \{0, 1\}$ indicates the current status of e_j , e.g. $\kappa = 1$ for active and $\kappa = 0$ for inactive. The maintenance cost is calculated with $q_j^m = \tau \cdot \kappa_r$, where τ is the unit cost for maintaining an active server.

In this way, the effective energy consumption considering only users can access edge web services can be expressed as:

$$\varrho_e(\mathcal{A}^t) = \sum_{u_j \in \Delta I, e_h \in \mathcal{E}} q_j^r + q_j^s + q_j^m \quad (19)$$

while the overall system energy consumption can be denoted by:

$$\varrho_t(\mathcal{A}^t) = \sum_{u_j \in \mathcal{U}, e_h \in \mathcal{E}} q_j^r + q_j^s + q_j^m \quad (20)$$

Algorithm 1: K-Anonymized Data Blurring (KADB).

1: **input:** users $\mathcal{U} = \{u_1, \dots, u_n\}$ with their original real location $\mathcal{X} = \{x_1, \dots, x_J\}$ and resource requests $\{y_1, \dots, y_J\} \in \mathcal{Y}$

2: **for** each user request $d_j^t = (x_j^t, y_j^t)$ **do**

3: calculate $\cos(d_j^t, d_o^t) \in \{\mathcal{D}\} \setminus \{a_j^t\}$ from u_j 's data a_j^t to all other user data

4: identify the k -nearest users with the shortest distances $\{d_1^{t'}, \dots, d_k^{t'}\} \in \mathcal{D}'$.

5: calculate obfuscated data $d_i^{t*} = \frac{1}{k} \sum_{d_o^{t'} \in \mathcal{D}'} d_o^{t'}$

6: **end for**

7: **output:** $\{d_1^{t*}, d_2^{t*}, \dots, d_J^{t*}\}$

The system energy efficiency $\Upsilon_e(\mathcal{A}^t)$ is determined by the ratio of the effective to the overall total energy consumption as follows:

$$\Upsilon_e(\mathcal{A}^t) = \frac{\rho_e(\mathcal{A}^t)}{\rho_t(\mathcal{A}^t)} \in [0, 1] \quad (21)$$

E. Privacy-Preserving Experience-Aware Edge Resource Rontrol

The objective of the PEERC problem is to formulate the optimal resource control function $a_{kn}(t)$ for users, to maximize the overall system welfare over time $[0, T]$.

Let $\Gamma(t, a_j^t, p_h(0))$ denote the overall system benefits of Edge demand allocation control in $t \in \mathcal{T}$ as the reward function of strategy $\mathcal{A}$. It consists of QoE, user coverage, user privacy, and energy efficiency with corresponding weights w_q, w_c, w_e, w_p ($w_q + w_c + w_e + w_p = 1$), represented as:

$$\begin{aligned} \Gamma(t, a_j^t, p_h(0)) = & w_q \cdot \sum_{u_j \in \mathcal{S}} \eta_j(a_j^t) + w_c \cdot \Upsilon_c(\mathcal{A}^t) + w_e \cdot \Upsilon_e(\mathcal{A}^t) \\ & + w_p \cdot \Upsilon_p(\mathcal{A}) - \sum_{e_h \in \mathcal{E}} \mathbf{1}_{\{p_h(t) < 0\}} * \Lambda \end{aligned}$$

where $\mathbf{1}_{\{p_h(t) < 0\}}$ represents a Dirac function, and Λ is a large enough constant serving as a harsh punishment. $p_h(0)$ is the initial resource of an edge machine e_h .

Therefore, we define the value function $\mathcal{V}_H(t, p_h(0))$ as the global resource control objective over time, expressed as:

$$\mathcal{O}_1 : \mathcal{V}_H(t, p_h(0)) = \sup_{\mathcal{A}^t \in [0, 1]^J} \frac{1}{H} \cdot \mathbb{E} \left[\int_0^T \Gamma(t, a_j^t, p_h(0)) dt \right]$$

With the value function, the optimal global resource control function is defined to maximize the overall system welfare.

IV. MEAN-FIELD EDGE RESOURCE ALLOCATION SYSTEM

In this section, we introduce a **mean-field edge resource allocation system (MERA)** to tackle the PEERC problem.

A. Service-Aware Data Masking

In web services, both user location and requested resources influence allocation efficiency. To protect these sensitive attributes, we consider cross-dimensional correlations and apply

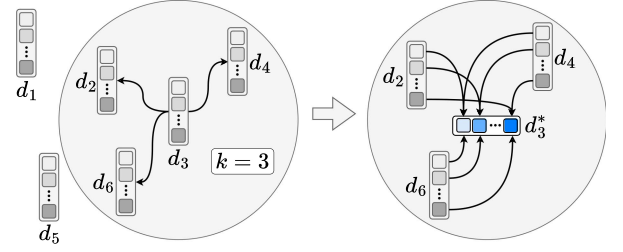


Fig. 4. Service-aware data obfuscation. Obfuscate user data $\{d_1, \dots, d_6\}$ with service-aware similarity-based k -anonymity, then forwards to the edge server for privacy-preserving allocation.

k -anonymity to generate obfuscated data for servers, which preserves user privacy while maintaining service efficiency.

In MERA, we introduce a novel algorithm, K -Anonymized Data Blurring (KADB), to protect user data against honest-but-curious edge machines on trustworthy base stations. KADB accounts for correlations across user request dimensions and leverages k -anonymity for privacy preservation. Before detailing KADB, we introduce a key metric for evaluating user privacy: cosine similarity between obfuscated and actual edge requests. As illustrated in Fig. 4, when $k = 3$, the nearest three neighbors are selected based on cosine similarity, and their information is combined to generate the obfuscated request. Higher cosine similarity indicates lower privacy, as it reflects closer alignment with the original data. The edge request cosine similarity is defined as:

Definition 6 (Edge Request Cosine Similarity): Consider user u_j 's request a_j^t and other user requests $a_o^{b,t}$, the edge request cosine similarity measures the similarity of is represented as follows:

$$\cos(d_j^t, d_o^t) = \frac{(x_j^t, y_j^{b,t}) \cdot (x_o^t, y_o^{b,t})}{|(x_j^t, y_j^{b,t})| |(x_o^t, y_o^{b,t})|} \quad (22)$$

The pseudo-code of KADB is presented in Algorithm 1, where we follow the notations that j indexes users and h indexes edge servers. Given user set $\mathcal{U}$ and request set $\mathcal{Y}$, KADB iteratively generates blurred data over time (Lines 2-7). In each iteration, it first computes the cosine similarity between edge requests of u_i and others u_o in $\mathcal{D}$ (Line 3). Then, the nearest k -neighbors of u_i are identified as $\mathcal{D}'$: $\mathcal{D}' = \{(x_o^{t'}, y_o^{t'}) \in \{\mathcal{X}, \mathcal{Y}\} \setminus \{x_j^t, y_j^t\} \mid \text{rank}(\cos(a_j^t, a_o^{t'}) \leq k)\}$. Next, the neighboring averaged location is used as the blurred data d_i^{t*} for user u_i (Line 5). This process continues until all requests are blurred, and the final blurred data uploaded for service provision is returned (Line 7), protecting users from potential attacks.

B. Mean-Field Learning for Resource Allocation

1) **Mean-Field Regularization:** As discussed in Section III-E, the value function $\mathcal{O}_1$ determines the optimal resource control strategies for each edge machine to maximize the system welfare. However, the dimension of joint interaction grows proportionally w.r.t. the number of users J , making it infeasible to learn individual strategies due to significant computational complexity with extensive information exchanges.

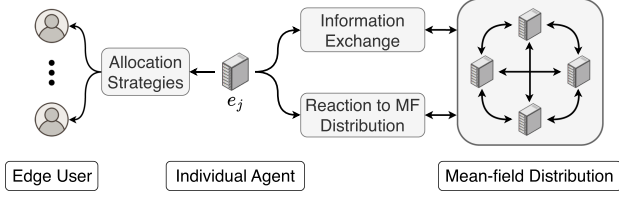


Fig. 5. Mean-field approximation. Edge users interact with individual server agents, exchanging information and strategies to form a mean-field distribution for collaborative control.

To address this, we regularize the problem using mean-field control with a global shared strategy to reduce complexity. Fig. 5 illustrates the mean-field approximation in the MEC system. Under this approximation, each server makes decisions based on the collective behavior of others represented as a unified distribution. Treating the remaining servers as an aggregated influence allows individual edge servers to collaboratively optimize resource allocation and improve overall system welfare.

Specifically, in a mean-field scenario, the remaining resources of edge machines can be formulated by a McKean-Vlasov process as:

$$-d\tilde{p}(t) = \sum_{u_j \in \mathcal{N}^t(e_h)} a_j(t, \mathcal{P}(t)) \mu_j \lambda_j dt + a_j(t, \mathcal{P}(t)) \lambda_j \sigma_j dW_t$$

$$\tilde{p}(0) = \frac{1}{H} \sum_{e_h \in \mathcal{E}} p_h(0)$$

where $\tilde{p}(0)$ is the initial value of representative particle in McKean-Vlasov dynamics. $\mathcal{P}(t)$ is the distribution of $\tilde{p}(t)$. Based on the derivation of QoE, user coverage, energy efficiency, and privacy in Section V-A, the mean-field reward function is given by:

$$\tilde{\Gamma}(t, a_j^t, \tilde{p}(0)) = w_q \cdot \sum_{u_j \in \mathcal{S}} \tilde{\eta}_j(a_j^t) + w_c \cdot \tilde{\Upsilon}_c(\mathcal{A}^t) + w_e \cdot \tilde{\Upsilon}_e(\mathcal{A}^t)$$

$$+ w_p \cdot \tilde{\Upsilon}_p(\mathcal{A}) - \sum_{e_h \in \mathcal{E}} \mathbf{1}_{\{\tilde{p}(t) < 0\}} * \Lambda$$

Therefore, the object $\mathcal{O}_2$ can be represented with proof in [24] as:

$$\mathcal{O}_2 : \tilde{\mathcal{V}}(t, \tilde{p}(0)) = \sup_{\mathcal{A}^t \in [0,1]^J} \mathbb{E} \left[\int_0^T \tilde{\Gamma}(t, a_j^t, \tilde{p}(0)) dt \right]$$

$$\text{where } \lim_{H \rightarrow \infty} \mathcal{V}_H(t, p_h^0) = \tilde{\mathcal{V}}(t, \tilde{p}(0)) \quad (23)$$

A global resource management strategy is obtained by solving this mean-field value function, allowing all servers to coordinate effectively and maximize system welfare.

C. Mean-Field Resource Q-Control

The dimension of joint action a can increase proportionally w.r.t the number of agents. This complexity makes it infeasible to handle the standard Q-function $Q_j(s, a)$ as each agent acts strategically and evaluates its value function based on the actions of all agents. To address this issue, we decompose the Q-function with mean-field theory by focusing solely on pairwise local

agent interactions, following [25], which can be represented as:

$$Q_j(\epsilon, a_j) = \frac{1}{\Omega_b(u_j)} \sum_{k \in \mathcal{N}_b(u_j)} Q_j(s, a_j, a_k) \quad (24)$$

where $\Omega_b(u_j)$ is the neighboring users size of u_j . While approximating interactions between an agent and its neighbors simplifies the multi-agent scenarios, it still captures the global interactions implicitly [26]. As illustrated in Fig. 5, compared with original MARL, the pairwise interaction $Q_j(s, a_j, a_k)$ in (24) can be further reduced by approximating others' effects with a virtual mean agent, which improves scalability and stability by capturing multi-agent interactions via aggregated population behavior.

Considering the group mean action $\bar{a}_j^h$ based on the neighborhood $\mathcal{N}(h)$ of edge machine e_h , then the one-hot action a_k^h of each neighbor k in terms of the sum of mean action $\bar{a}_j^h$ with a small fluctuation $\delta \cdot a_j^k$ as (superscript h is omitted from $a_j^{h,k}$ for simplicity):

$$a_j^k = \bar{a}_j^h + \delta \cdot a_j^k, \quad \text{where } \bar{a}_j^h = \frac{1}{\Omega_h} \sum_k a_j^k \quad (25)$$

The mean-field Q-function can be approximated by an additive decomposition and Taylor's expansion:

$$Q_h(s, a) = \frac{1}{\Omega_h} \sum_k Q_h(s, a_j^h, a_j^k)$$

$$= Q_h(s, a_j, \bar{a}_j) + \nabla_{a_j} Q_h(s, a_j, \bar{a}_j) \cdot \left[\frac{1}{\Omega_h} \sum_k \delta a_j^{h,k} \right]$$

$$+ \frac{1}{2\Omega_h} \sum_k \left[\delta a_{j,k} \cdot \nabla_{a_j,k}^2 Q_h(s, a_j, \bar{a}_j) \cdot \delta a_j^{h,k} \right]$$

$$= Q_h(s, a_j, \bar{a}_j) + \frac{1}{2\Omega_h} \sum_k R_{s,a_j,j}(a^k) \approx Q_h(s, a_j, \bar{a}_j)$$

where $R_{s,a_j,j}(a_j^k) \triangleq \delta a_j^{h,k} \cdot \nabla_{a_j,k}^2 Q_h(s, a_j, \bar{a}_j) \cdot \delta a_j^{h,k}$ denotes the Taylor polynomial's remainder with $\bar{a}_j^{h,k} = \bar{a}_j + \epsilon_j^{h,k} \delta a_j^{h,k}$ and $\epsilon_j^{h,k} \in [0, 1]$. Detailed derivation can be found in Section V-A.

Remark: This expansion approximates the influence of neighboring agents by expanding the Q-function around the mean action $\bar{a}_j$, capturing their collective effect without modeling all pairwise interactions. The first-order term reflects deviations from the mean, while the higher-order remainder represents smaller, bounded interactions. Thus, the Q-function can be well approximated by its mean-field form $Q_h(s, a_j, \bar{a}_j)$, significantly reducing computational complexity.

With the mean-field approximation, the pairwise interaction $Q_h(s, a_j^h, a_j^k)$ between agent e_h and neighboring agent e_k is simplified as between the central agent and the virtual mean agent, representing the averaged behavior of all neighbors nearby e_h , i.e., $Q_h(s, a_j^h, \bar{a}_j^h)$. The mean-field Q-function is recurrently updated as:

$$Q_h^{t+1}(s, a_j, \bar{a}_j) = (1 - \alpha) Q_h^t(s, a_j, \bar{a}_j) + \alpha [\Gamma + \gamma \mathcal{V}_h(s')] \quad (26)$$

Algorithm 2: Mean-Field Resource Q-Control.

```

1: initialization
2: Retrieve user requests by Algorithm 1
3: Initialize available resources for all servers, the weights
  of Q-functions  $Q_h$  and the mean-field action  $\bar{a}_j$  for all
  users  $u_j \in \mathcal{S}$ 
4: Initialize the total steps, episodes and replay buffer  $RB$ 
5: end initialization
6: for each episode do
7:   Update parameters from PEERC environment.
8:   for each step do
9:     Sample  $a_j$  from mean-field q-policy induced by
       $Q_h$  according to (29) current mean action  $\bar{a}_j$  and
      exploration ratio  $\beta$ 
10:    Compute the new mean action  $\bar{a}_j^t$ 
11:    Execute the joint allocation strategy
       $\{a_1, \dots, a_J\} \in \mathcal{A}$ , observe rewards, and the next
      state
12:    Store the experiences  $\langle s, a, r, s', \bar{a}_j^t \rangle$  in the  $RB$ 
      where  $\{\bar{a}_1^t, \dots, \bar{a}_J^t\} \in \bar{\mathcal{A}}$  is the mean action
13:    for user  $j = 1$  to  $J$  do
14:      Sample a minibatch of experiences from  $RB$ .
15:      Set  $y_j$  according to (27).
16:      Update Resource critic/actor network by
        minimizing the loss in (30) and (31) averaged by
        the samples.
17:    end for
18:    Update params of target networks with (32) and
      (33)
19:  end for
20: end for
21: output Optimal allocation strategies  $\{a_1^{t*}, a_2^{t*}, \dots, a_J^{t*}\}$ 

```

where $\gamma \in [0, 1]$ is the reward discount factor, and α denotes the learning rate. The mean-field value function is:

$$\tilde{V}_h^t(s') = \sum_{a_j} \pi_j(a_j | s', \bar{a}_j) \cdot \mathbb{E}[Q_h(s', a_j, \bar{a}_j)] \quad (27)$$

In this case, the mean-field approximation reduces the multi-agent PEERC problem to finding the optimal response strategy for the central server e_h with respect to the average action $\bar{a}_j^h$ of its neighboring agents, greatly simplifying real-time implementation. The mean action $\bar{a}_j$ and optimal EDR policy can then be expressed with mean actions $\bar{a}_j^{k-}$ as:

$$\bar{a}_j^h = \frac{1}{\Omega_h} \sum_{k \neq h} a_j^k, \quad a_j^k \sim \pi_k(\cdot | s, \bar{a}_j^{k-}) \quad (28)$$

$$\pi_h^t(a_j | s, \bar{a}_j) = \frac{\exp(-\beta Q_h(s, a_j, \bar{a}_j))}{\sum_{a_{j'} \in \mathcal{A}_j} \exp(-\beta Q_h(s, a_{j'}, \bar{a}_j))} \quad (29)$$

By (28) and (29), the mean action $\bar{a}_j$ and π_h^t updates iteratively. Next, we implement the mean-field Q-function in (26) with the actor-critic network under continuous spaces [27]. The agent e_h is trained by minimizing the loss function for the critic network:

$$\mathcal{L}_a(\theta_h^c) = (y_h - Q_{\theta_h^c}(s, a_j, \bar{a}_j))^2 \quad (30)$$

differentiated as $\nabla_{\theta_h^c} \mathcal{L}_a(\theta_h^c) = Q_{\theta_h^c}(s, a_j, \bar{a}_j) \nabla_{\theta_h^c} Q_{\theta_h^c}(s, a_j, \bar{a}_j)$. The actor (policy) network is trained by:

$$\nabla_{\theta_h^p} \mathcal{L}_c(\theta_h^p) \approx \nabla_{\theta_h^p} \log \pi_{\theta_h^p}(s) Q_{\theta_h^c}(s, a_j, \bar{a}_j) \Big|_{a=\pi_{\theta_h^p}(s)} \quad (31)$$

Given the above optimizations, the parameters of the shared critic θ^c and actor θ^p are updated iteratively until convergence as follows:

$$\theta_h^{c-} \leftarrow \alpha_c \cdot \theta_h^c + (1 - \alpha_c) \cdot \theta_h^{c-} \quad (32)$$

$$\theta_h^{p-} \leftarrow \alpha_p \cdot \theta_h^p + (1 - \alpha_p) \cdot \theta_h^{p-} \quad (33)$$

The pseudocode of the actor-critic mean-field resource Q-control is represented in Algorithm 2. It starts with initializations (Lines 1-5). During each episode, the environment parameters are updated, and actions are sampled based on the current policy (Lines 7-9). Then it computes the new mean actions and executes the joint state while storing experiences in the replay buffer (Lines 10-12). Using minibatch samples, it updates both the resource critic and actor networks, along with the target networks (Lines 14-18). Finally, it produces the optimal allocation strategies (Line 21).

V. THEORETICAL ANALYSIS

A. Mean-Field Approximation

We present the transformation from n -particle dynamics to McKean–Vlasov dynamics to formulate the mean-field control problem. In the MEC system, resource allocation across multiple edge servers is modeled as a multi-agent collaboration. Using the mean-field approximation, we replace individual interactions with the population average. Following Definition 2 and Section IV-B1, we derive the mean-field transformation for random variables in the multi-agent PEERC environment.

The QoE can be reformulated in the mean-field scenario as:

$$\tilde{\eta}_j(a_j^t) = \frac{\zeta}{1 + e^{-\delta(\tilde{i}(a_j^t) - \gamma)}} \quad (34)$$

where $\tilde{i}(a_j^t)$ approximated as:

$$\tilde{i}(a_j^t) = \frac{1}{1 - a_j^t(\tilde{p}(t), x_j)} - \frac{1}{a_j^t(\tilde{p}(t), x_j)} \quad (35)$$

where $\tilde{p}(t)$ refers to the aggregate impact of all other agents at time t . The sigmoid function approximates how QoE saturates with the increasing resource allocation and mean-field interaction effects. And, the system's overall user coverage can be approximated over time $\mathcal{T}$, which is represented as:

$$\tilde{\mathcal{F}}_{total}(\mathcal{A}^t) = \frac{1}{\mathcal{T}} \cdot \sum_{t \in \mathcal{T}} \mathcal{F}_j(\mathcal{A}^t) \quad (36)$$

The mean-field user coverage is then defined as:

$$\tilde{\Upsilon}_c(\mathcal{A}^t) = \frac{\tilde{\mathcal{F}}_{total}}{J} \in [0, 1] \quad (37)$$

where J is the total number of users in the system, and $\tilde{\Upsilon}_c(\mathcal{A})^t$ is the normalized user coverage.

The system's energy efficiency can be decomposed into resource consumption components such as the running cost q_j^r , switching cost q_j^s , and the maintenance cost q_j^m .

For the effective users in ΔI , the mean-field energy consumption for resource allocation is computed as:

$$\tilde{q}_e(\mathcal{A}^t) = \sum_{u_j \in \Delta I} q_j^r + q_j^s + q_j^m \quad (38)$$

while for the total energy consumption for all users $\mathcal{U}$ is denoted by:

$$\tilde{q}_t(\mathcal{A}^t) = \sum_{u_j \in \mathcal{U}} q_j^r + q_j^s + q_j^m \quad (39)$$

Finally, the mean-field energy efficiency across the system is defined as the ratio of resource consumption by effective users to total consumption, formulated as:

$$\tilde{\gamma}_e(\mathcal{A}^t) = \frac{\tilde{q}_e(\mathcal{A}^t)}{\tilde{q}_t(\mathcal{A}^t)} \in [0, 1] \quad (40)$$

The data privacy can be further approximated by:

$$\tilde{\psi}_j^l = \sum_{x_j \in \mathcal{X}} \mathcal{J}(\hat{x}_j | x_j^*) \cdot \|x_j - \hat{x}_j\| \text{ and } \tilde{\psi}_j^r = \left\| \frac{y_j^* - y_j}{y_j} \right\|_2^2 \quad (41)$$

Then individual data privacy is represented as:

$$\tilde{\mathcal{R}}_j = w_1 \cdot \left\| \frac{\tilde{\psi}_j^l}{f_j} \right\|_2^2 + w_2 \cdot \tilde{\psi}_j^r \quad (42)$$

Therefore, the overall system privacy under the mean-field scenario can be formulated as:

$$\tilde{\gamma}_p(\mathcal{A}^t) = \frac{1}{\mathcal{T}} \cdot \sum_{u_j \in \Delta I, t \in \mathcal{T}} \frac{\tilde{\mathcal{R}}_j}{\mathcal{M}(\mathcal{A}^t)} \in [0, 1] \quad (43)$$

The mean-field approximation in the multi-agent PEERC problem enables the reduction of complex interactions between individual servers and the interaction of each server with the aggregate behavior of all other servers for real-time service delivery. Additionally, we have also proved the convergence of the mean-field approximation in Appendix B, available online.

VI. EVALUATION

This section presents extensive experiments to evaluate the performance of MERA. The dataset and code are available for reproduction at <https://github.com/ziquwang20/MERA>.

A. Experiment Settings

Experimental Data: We conduct experiments using the EUA edge resource allocation dataset [28] with server features from AWS Wavelength [29] and Alibaba Cloud [30] to evaluate our approach in various MEC scenarios. The synthetic dataset (publicly available in our repo) contains heterogeneous user resource requests and edge server configurations. User-side data include task demands characterized by resource requirements (e.g., computation and bandwidth) and spatial-temporal variations, while server-side data provide information on capacity, coverage, and energy-related attributes.

Implementation: For model training, we conducted experiments using the edge resource actor-critic algorithm over 10000

TABLE II
SYSTEM SETTINGS

	Users J	Edge Servers H	Privacy k
Set #1	100, 150, 200, 250, 300	60	6
Set #2	200	40, 50, 60, 70, 80	6
Set #3	200	60	2, 4, 6, 8, 10

episodes, each with a maximum trajectory length of 10000 steps. Exploration noise followed a Gaussian distribution $\mathcal{N}(0, 0.1)$ to encourage exploration. The actor and critic networks were optimized using Adam with learning rates of 10^{-4} and 10^{-3} , respectively. A soft target update was applied with $\tau = 10^{-2}$. The exploration rate is initialized with Gaussian noise $\mathcal{N}(0, 0.1)$ to promote early-stage exploration.

The experiments assess the impact of three key parameters: 1) the number of users J , 2) the number of edge servers H , and 3) the privacy parameter k , with each parameter varied in Table II to analyze its effect across various metrics on system welfare. We conduct additional experiments to evaluate MERA under diverse real-world scenarios, including varying QoE thresholds, system coefficients, and small-scale settings. We further provide case studies on small-scale scenarios (Section VII) to demonstrate system robustness, as well as ablation studies comparing different protection methods (especially k -anonymity and DP) to highlight component effectiveness and real-world applicability (Section VIII).

Parameter Guidelines: In practice, the key parameters in MERA can be configured to meet different deployment needs. The privacy parameter k controls data indistinguishability, where a larger k enhances privacy at the cost of slight performance degradation. The QoE threshold ι defines the minimum acceptable service quality, with higher values improving user experience but potentially reducing system coverage. The system weights $\mathbf{W}$ balance multiple objectives, including QoE, energy efficiency, and privacy, enabling different operational priorities. Overall, these parameters can be adjusted to balance privacy, performance, and efficiency.

B. Benchmarks

To evaluate MERA, we compare it with the state-of-the-art privacy- and energy-preserving systems GEES [23] and EESaver [31]. GEES uses differential geo-obfuscation with energy-aware allocation, while EESaver reduces energy use by powering off idle servers. For a fair privacy comparison, we additionally implement Laplace [32] (EES-L) and Gaussian [33] (EES-G) mechanisms for EESaver with $\sigma = \ln 6$. The detailed definitions can be found in Appendix D, available online.

C. System Performance

Table III summarizes system performance across system metrics for experiment Sets #1-3. MERA consistently achieves the best overall results among all benchmarks.

1) Impact of User Numbers: System scalability is crucial for real-world edge web applications, ensuring that MERA maintains strong performance as the user population grows. When users increase from 100 to 300 in Set #1, MERA exhibits a

TABLE III
EXPERIMENTAL RESULTS ON VARIOUS METRICS

Metrics	Benchmarks	# of Users					# of Edge Servers					k -anonymity / $\sigma(\ln)$				
		100	150	200	250	300	40	50	60	70	80	2	4	6	8	10
Overall	MERA	73.55	71.13	70.26	69.10	67.12	67.83	68.75	70.26	70.59	70.59	70.35	69.48	70.26	70.09	68.27
	GEES	58.05	55.96	55.14	53.00	52.43	52.85	52.94	55.14	55.73	56.65	53.58	54.61	55.14	60.88	62.36
	EES-L	64.25	61.96	60.60	60.49	55.13	57.67	59.44	60.60	65.22	70.51	59.61	60.56	60.60	65.11	68.11
	EES-G	53.60	52.23	51.37	49.67	46.52	48.40	50.61	51.37	52.17	53.50	47.57	50.72	51.37	52.70	55.53
QoE	MERA	82.76	81.32	80.65	78.06	75.18	76.04	77.48	80.65	81.37	82.03	81.23	79.91	80.65	80.45	78.61
	GEES	69.00	66.00	65.00	62.00	61.00	60.00	61.00	65.00	66.50	69.50	62.00	63.50	65.00	72.00	74.50
	EES-L	81.00	78.00	76.50	76.00	68.00	72.00	74.50	76.50	82.00	89.00	73.50	75.50	76.50	82.00	86.50
	EES-G	68.00	66.00	64.50	62.00	58.00	60.00	63.00	64.50	65.50	67.50	58.50	63.00	64.50	66.00	69.50
Coverage	MERA	95.00	88.00	85.00	84.00	79.00	80.00	82.00	85.00	86.50	87.50	86.00	84.00	85.00	86.50	82.00
	GEES	71.00	68.00	67.00	64.00	63.00	62.00	63.00	67.00	68.50	69.50	64.00	65.50	67.00	74.00	76.50
	EES-L	81.00	78.00	76.50	76.00	68.00	72.00	74.50	76.50	82.00	89.00	73.50	75.50	76.50	82.00	86.50
	EES-G	68.00	66.00	64.50	62.00	58.00	60.00	63.00	64.50	65.50	67.50	58.50	63.00	64.50	66.00	69.50
Privacy	MERA	23.16	25.83	28.57	30.06	31.45	24.54	28.43	28.57	29.14	29.69	24.57	26.88	28.57	27.98	29.82
	GEES	21.32	21.61	21.88	21.14	21.04	23.12	22.31	21.88	23.30	23.70	23.76	23.44	23.30	22.41	21.58
	EES-L	13.15	13.20	12.06	13.82	14.40	13.68	13.43	12.06	14.01	14.48	17.64	14.47	13.06	13.27	13.21
	EES-G	10.09	10.30	11.18	11.34	11.80	11.76	12.58	11.18	10.95	10.88	13.85	12.21	11.18	11.39	11.71
Energy	MERA	93.28	89.37	86.81	84.26	82.86	90.73	87.08	86.81	85.33	83.12	89.61	87.12	86.81	85.43	82.63
	GEES	70.87	68.24	66.69	64.85	64.70	66.29	65.43	66.69	64.62	63.91	64.58	65.98	66.69	75.12	76.87
	EES-L	81.84	78.62	77.32	76.14	70.11	72.98	75.33	77.32	82.87	89.62	73.79	76.78	77.32	83.16	86.22
	EES-G	68.29	66.62	65.28	63.32	58.26	61.85	63.85	65.28	66.74	68.11	59.42	64.67	65.28	67.42	71.39

steady improvement in data privacy with only slight declines in QoE and coverage. Specifically, privacy increases from 23.16 to 31.45, while MERA still achieves the highest QoE, user coverage, and energy efficiency, averaging 79.59, 86.20, and 87.32. These trends reflect increased resource contention under larger user populations, which slightly affects service quality. In contrast to benchmarks with largely unchanged privacy, MERA continues to enhance privacy due to the increased candidate pool for k -anonymity. Although slightly lower than GEES in privacy, MERA remains competitive and consistently outperforms all other baselines in QoE, coverage, and energy efficiency, demonstrating strong scalability and robustness.

2) *Impact of Edge Machine Numbers*: Increasing the number of edge servers from 40 to 80 (Set #2) leads to notable improvements in service performance with stable data privacy. For MERA, QoE and user coverage increase from 76.04 and 80.00 to 82.03 and 87.50, while energy efficiency decreases from 90.73 to 83.12. The added computational capacity enables more efficient responses to user requests, reducing resource competition and improving service quality. Similar trends appear in the benchmarks: QoE and coverage for GEES, EES-L, and EES-G increase from 63.00, 74.50, and 63.00 to 69.50, 89.00, and 67.50. However, MERA consistently optimizes the additional server capacity, achieving stronger overall performance. Notably, MERA delivers higher data privacy and system performance as the number of servers grows, highlighting its superior resource management and privacy-preserving capabilities.

3) *Impact of Privacy Level*: As the k -anonymity value increases from 2 to 10 (corresponding to the DP noise budget σ for benchmarks) in Set #3, a clear trade-off emerges between data privacy and service performance. For MERA, QoE remains relatively stable as k grows, while user coverage and energy efficiency show a slight decline. In contrast, data privacy rises significantly from 24.57 to 29.82, since higher k levels introduce greater data distortion, reducing QoE and adding computational overhead for accurate service delivery. For GEES, EES-L, and

TABLE IV
CASE EXPERIMENT SETTINGS

	QoE Threshold ι	System Coeff. $\mathcal{W}$	Edge Servers H	Users J
Set #4	0.6, 0.7, 0.8, 0.9	$\mathcal{W}_1$	60	200
Set #5	0.7	$\mathcal{W}_1, \mathcal{W}_2, \mathcal{W}_3, \mathcal{W}_4$	60	200
Set #6	0.7	$\mathcal{W}_1$	2, 6, 10	9
Set #7	0.7	$\mathcal{W}_1$	6	3, 9, 15

$\mathcal{W}_1 = (0.25, 0.25, 0.25, 0.25)$, $\mathcal{W}_2 = (0.3, 0.3, 0.2, 0.2)$, $\mathcal{W}_3 = (0.10, 0.10, 0.40, 0.40)$, $\mathcal{W}_4 = (0.40, 0.30, 0.20, 0.10)$.

EES-G, data privacy decreases from 47.52, 35.28, and 27.69 to 43.15, 26.41, and 23.41, which leads to improved QoE, coverage, and energy efficiency. This occurs because, in differential privacy, a larger σ indicates weaker privacy protection. As σ increases, privacy constraints are gradually relaxed, resulting in higher system performance. These observations further suggest that, compared to noise-based mechanisms such as differential privacy, MERA's k -anonymity better preserves demand structure and maintains stable service performance, making it more suitable for latency-sensitive MEC scenarios.

In summary, MERA consistently outperforms the benchmarks in overall system welfare. It achieves a strong balance between privacy protection and performance, maintaining high QoE and coverage while improving privacy. These results highlight MERA's robust performance and its adaptability and scalability across diverse edge scenarios.

VII. CASE STUDY ON SMALL SCALE SCENARIO

A. Case Experiment Settings

Table IV presents the experimental settings for the case studies, including variations in Set #4 (number of users), Set #5 (edge servers), and Set #6 (system coefficients). The system coefficient $\mathcal{W}$ is defined with three configurations based on the preferences of service providers.

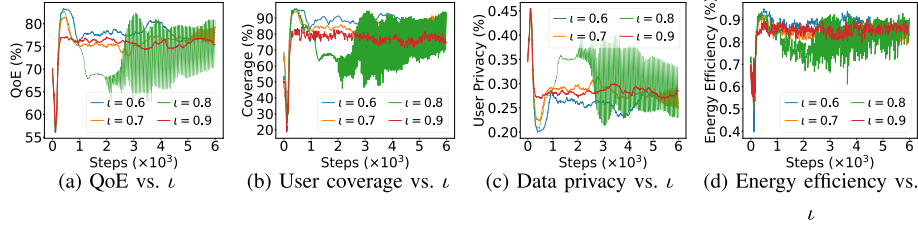


Fig. 6. Case study set #4: Variations on QoE threshold.

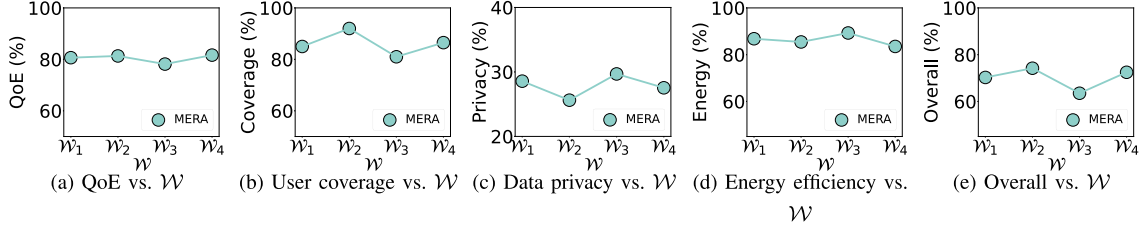


Fig. 7. Case study set #5: Variations on system coefficients.

B. Variations on QoE Thresholds

The QoE threshold ι determines whether a user received satisfied enough service to include her for coverage, i.e., if the QoE exceeds the ι , the user is counted in the system user coverage. We examine our proposed method MERA of this impact by varying ι from 0.6 to 0.9 over 6000 steps.

Fig. 6 presents the experimental results for QoE, user privacy, user coverage, and energy efficiency. As shown in Section VII-A, at lower thresholds ($\iota = 0.6$), QoE stabilizes at a relatively high level, whereas higher thresholds exhibit greater fluctuations. User coverage in Section VII-A decreases as ι increases: at $\iota = 0.6$ more users remain within system coverage, while coverage drops sharply at $\iota = 0.8$ and 0.9 because fewer users satisfy the stricter QoE requirement. This highlights the trade-off between QoE strictness and coverage. As shown in Section VII-A, user privacy generally improves with higher ι , though at the cost of reduced user satisfaction. Meanwhile, Section VII-A indicates that energy efficiency remains largely stable across all thresholds, with MERA consistently maintaining around 90

C. Variations on System Coefficients

Depending on operational goals and constraints, service providers may prioritize different objectives such as QoE, energy efficiency, or privacy. To evaluate MERA's adaptability under varying preferences, we examine multiple system coefficient settings. As shown in Fig. 7, MERA exhibits distinct trade-offs across $\mathcal{W}$ configurations. The default $\mathcal{W}_1$ achieves balanced performance, while $\mathcal{W}_2$ attains the highest overall welfare of 74.21%, supported by strong QoE (81.32%) and user coverage (92%) at the cost of reduced privacy. The $\mathcal{W}_3$ setting improves energy efficiency by prioritizing coverage and utilization, but yields the lowest overall performance. In contrast, the QoE-focused $\mathcal{W}_4$ achieves the highest QoE (81.65%) while maintaining a reasonable balance across other metrics. Overall, these

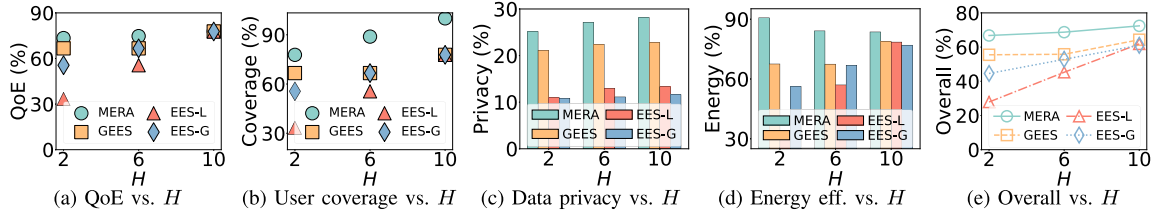
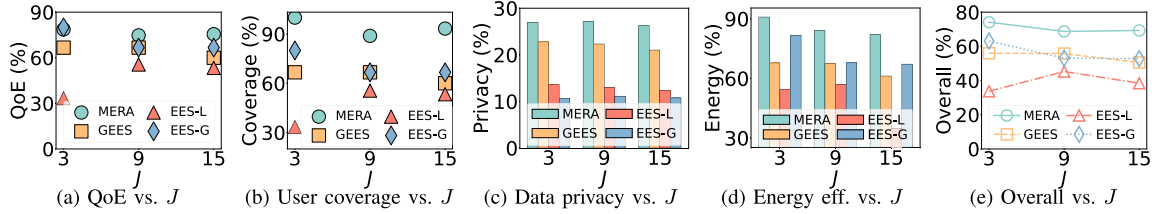
results demonstrate that MERA can effectively adapt to different priority settings, achieving flexible trade-offs and robust system performance across diverse scenarios.

D. Performance Under Small Scale Scenarios

In edge services, improving user privacy and system performance requires adaptability to different user and server scales. For service providers, scalable and flexible solutions are essential to maintain efficiency, reliability, and QoS across varied deployment scenarios. To evaluate MERA under smaller-scale conditions, we conducted a case study with reduced user and server populations.

Impact on Server Numbers: As shown in Fig. 8, with the expansion of edge servers from 2 to 10, MERA consistently delivers superior performance, effectively balancing system welfare. With more servers, the users' QoE rises from 73.44% to 77.86%, while data privacy improves from 25.17% to 28.18% with 100% user coverage, ensuring seamless service quality. Although the system energy efficiency sees a slight decline, MERA still maintains a well-optimized trade-off across all metrics. In contrast, benchmark systems of GEES, EES-L, and EES-G exhibit instability, with weaker data privacy protection and fluctuating user coverage, which seriously reduces the system's utility. These findings confirm MERA's scalability, showing its ability to sustain high performance with privacy guarantees in small-scale edge scenarios.

Impact on User Numbers: As user numbers increase in small scale from 3, 6, 9, Fig. 9 demonstrates that MERA consistently outperforms benchmark methods in both individual metrics and overall system efficiency. Notably, the user coverage improves from 83.33% to 93.33%, while the QoE remains stable around 74–79%, and system energy efficiency stays above 82%, ensuring a balanced trade-off across key performance indicators. In contrast, GEES, EES-L, and EES-G exhibit noticeable declines

Fig. 8. Small scale set #6: Variations on server H .Fig. 9. Small scale set #7: Variations on users J .

in QoE, data privacy, and user coverage, with overall performance dropping as user demand grows. These results highlight MERA's adaptability and robustness, effectively maintaining high QoE, Privacy, and system-wide efficiency even as the system scales.

VIII. ABLATION STUDY

To evaluate the contributions of individual components in the MERA system and the impact of various configurations on its performance, we present a comprehensive ablation study in this section. By systematically removing and adjusting the components in MERA, we quantify their effects on system welfare to demonstrate the effectiveness.

A. Ablation Settings

We evaluate four scenarios to isolate the effects of MERA's components and configurations. Each scenario focuses on a specific aspect of the system under different real-world conditions, using metrics such as QoE, user coverage, data privacy, and Energy eff.. The scenarios are as follows:

- *MERA with semi-privacy protection.* In this scenario, the KADB module is disabled, and the protection in MERA only considers the reward function, using raw user data for resource allocation.
- *MERA without privacy protection* Here we fully remove both privacy components in MERA: KADB and the privacy-aware reward function in MFRQ, to evaluate the resource allocation performance under no privacy protection scenarios.
- *MERA's k -anonymity KADB with benchmarks* We further investigate the performance using our KADB combining with other benchmark resource allocation methods to demonstrate the effectiveness of our MFRQ.

TABLE V
ABLATION STUDY SETTINGS

	Users J	Edge Servers H	Privacy k
Set #8	100,200,300	60	Semi
Set #9	200	40,60,80	Semi
Set #10	100,200,300	60	NA
Set #11	200	40,60,80	NA
Set #12	200	60	2,4,6,8

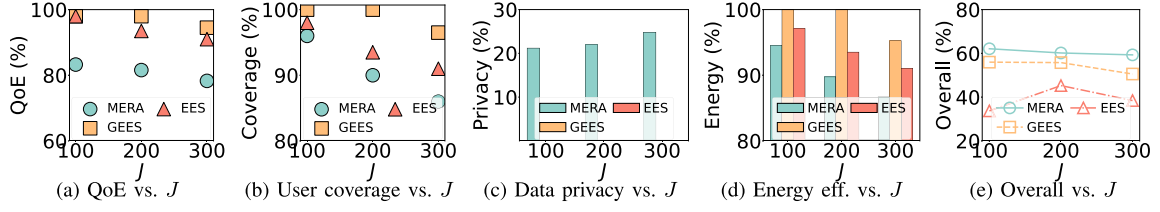
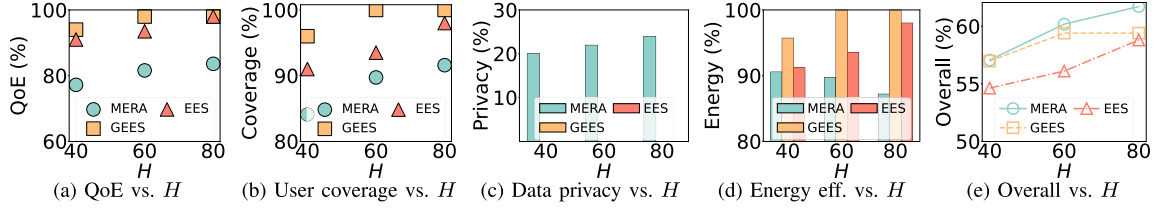
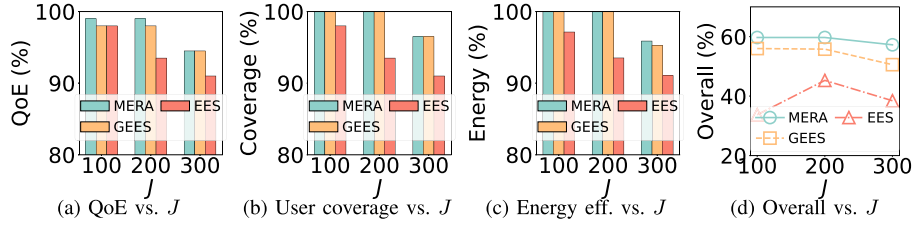
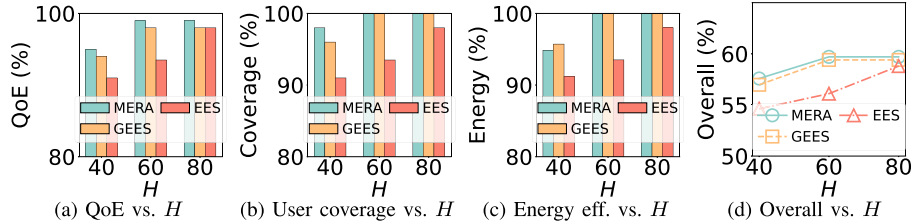
Based on the above scenarios, the experiment settings are detailed in Table V.

B. Ablation System Performance

We evaluate the performance of MERA according to the above scenarios, including variations in user (100, 200, 300) and server numbers (40, 60, 80).

1) *MERA With Semi-Privacy Protection:* As shown in Fig. 10, MERA consistently achieves the highest overall system welfare at 62.11%, 60.18%, and 59.31%, outperforming GEES and EESaver. It also provides notable privacy protection (e.g., 24.83% at 300 users), whereas GEES and EESaver offer none in this setting. QoE, energy efficiency, and coverage remain competitive, with MERA performing on par with GEES and better than EES-L while delivering additional privacy. Similarly, as the number of servers increases (Fig. 11), MERA maintains the best overall performance, achieving welfare scores of 57.06%, 60.18%, and 61.71%. It also preserves strong privacy and user coverage compared to the benchmarks. This balanced performance and strong privacy guarantees make MERA the most effective solution in privacy-sensitive scenarios.

2) *MERA Without Privacy Protection:* As shown in Fig. 12, MERA achieves the highest overall system welfare at 59.70%, 59.70%, and 57.24%, outperforming GEES (53.22%) and

Fig. 10. Ablation set 8: Variations on users J (semi-privacy).Fig. 11. Ablation set 9: Variations on servers H (semi-privacy).Fig. 12. Ablation set 10: Variations on users J (without privacy).Fig. 13. Ablation set 11: Variations on servers H (without privacy).

EESaver (39.15%) on average. Under no privacy-protection settings, MERA also delivers top QoE performance, reaching 94.5% for 300 users, matching GEES and surpassing EESaver's 91%. Its energy efficiency remains strong at 95.88%, comparable to GEES (95.27%) and better than EESaver (91.07%). As server numbers increase (Fig. 13), MERA maintains the highest welfare and continues to lead in system performance, demonstrating superior performance and robustness even excluding privacy modules.

3) *MERA's k -Anonymity KADB With Benchmarks:* Fig. 14 demonstrates the performance of MERA's KADB combined with other benchmark methods. As the privacy k -anonymity values of k increase from 2, 4, 6, to 8, MERA consistently achieves the highest overall system welfare, significantly outperforming GEES (54.38%, 52.69%, 49.43%, 48.38%) and EESL

(48.53%, 48.02%, 46.89%, 45.77%). MERA excels in privacy, reaching 28.57% at $k = 6$, far exceeding GEES (13.65%) and EESaver 15.12%. Also, MERA's QoE remains stable across all settings, e.g., 81.23% at $k = 2$, higher than GEES (80.5%) and EESaver (71%). This further demonstrates MERA's superior ability in privacy preservation while maintaining competitive performance.

IX. ADDITIONAL EVALUATION

To further extend the evaluation of MERA, we include additional comparisons with two recent privacy-aware MEC approaches as supplementary benchmarks to provide broader empirical insights, which are MADDPG [34] and DQN-DP [35] (see Table VII in Appendix), available online.

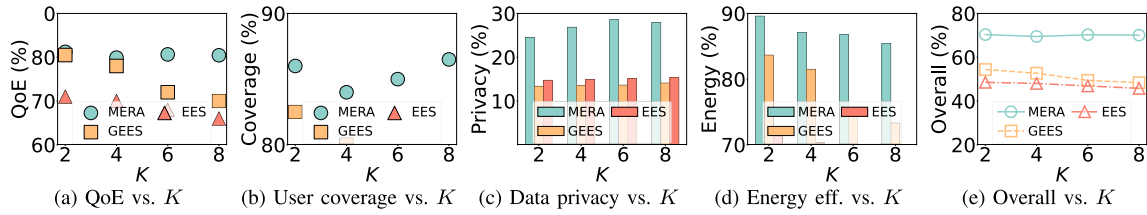


Fig. 14. Ablation set 12: Variations on privacy level k (k -anonymity based benchmarks).

Notably, they focus on a subset of the PEERC problem, whereas our proposed MERA and the other benchmarks in the main text focus on the PEERC problem from multiple metrics perspectives. Specifically, [34] focuses on auction-based resource allocation with decentralized coordination, while [35] adopts differential privacy for location protection under relatively static settings.

Across all settings, MERA consistently achieves the best performance with a clear margin over both benchmarks. Under varying user scales, MERA attains an overall score of 73.55–67.12, outperforming them by approximately 30–40%, while maintaining higher QoE (e.g., 82.76 vs. 68.20/64.80) and coverage (95.00 vs. 70.50/66.00). It also provides stronger privacy guarantees (e.g., 23.16 vs. ~ 10 –11) and higher energy efficiency across all configurations. Overall, these results demonstrate that MERA achieves a more balanced and robust trade-off across multiple objectives.

X. RELATED WORK

A. Edge Demand Response

The rise of multi-access edge computing (MEC) has spurred research in areas like edge-assisted learning [36], [37], mobile crowdsensing [38], [39], and edge data privacy [40], [41]. However, optimizing resource allocation in dynamic MEC environments remains challenging due to balancing real-time service delivery, computational overhead, privacy, and energy efficiency. To address this challenge, Chen et al. [42] introduced an energy optimization mechanism using online auctions; however, its cloud-centric focus limits scalability. Edgedr [22] made efforts by enabling flexible server energy control, but it lacks consideration for user privacy and falls short in service latency. Cui et al. [31] later proposed EESaver, which enhances performance while reducing energy consumption, though it struggles to address growing privacy concerns as well. GEES [23], as the state-of-the-art framework in privacy preservation, offers robust theoretical guarantees but compromises the quality of experience and user coverage, incurring significant computational delays. Thus, developing an EDR framework that enables real-time service delivery with satisfying user coverage and privacy guarantees in MEC remains a critical challenge.

B. Multi-Agent Reinforcement Learning

Multi-agent reinforcement learning (MARL) was first developed using independent Q-learning, where each agent learns

without considering others [43]. However, this often fails due to the non-stationary nature of multi-agent environments [44]. To address this, approaches such as shared policy parameters [4], incorporating other agents' policies [45], and importance sampling [46] have been proposed. More recently, centralized training with decentralized execution, combined with policy gradients [47] and actor-critic frameworks [48], has shown strong potential by allowing agents to learn a centralized Q-function during training while acting locally at execution. Yet these methods typically scale poorly to large agent populations [19]. To improve scalability, mean-field reinforcement learning applies mean-field approximations to extend MARL to large populations [25], achieving proven convergence for tabular Q-functions. Building on this, our work further extends MARL to Q-function approximators, ensuring convergence in more complex settings.

XI. CONCLUSION

This paper investigates the privacy-preserving experience-aware edge resource control (PEERC) problem in MEC environments. To tackle the PEERC problem, we propose a global edge resource allocation system named MERA to maximize overall system welfare with privacy guarantees. MERA leverages k -anonymity to protect user privacy while maximizing overall system performance across various metrics with collaborative mean-field reinforcement learning. Extensive experiments demonstrate that MERA consistently outperforms representative approaches, establishing it as a state-of-the-art solution for edge resource allocation. In the future, we will explore safeguarding a broader range of users' sensitive information and delve into more advanced attack-defense mechanisms for MERA.

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